Criteria

Knowledge units are categorized to:

- Essential units
- Core units
- Specialized units

Class Groups

Essential Units

Two (2) units:

- GKU 0.1 - Linear Algebra
- GKU 0.2 - Statistics and Probability

Core Units

Nine (9) units:

- GKU 1 - Algorithms, Computability, and Complexity
- GKU 2 - Architecture and Organization
- GKU 3 - Discrete Structures
- GKU 4 - Information Management
- GKU 5 - Networking and Communications
- GKU 6 - Operating Systems
- GKU 7 - Programming Languages
- GKU 8 - Software Development Fundamentals
- GKU 9 - Software Engineering

Specialized Knowledge Units (SKU)

A list of SKUs for each GKUs.

0.1. Linear Algebra

- Matrices
- Systems of equations
- Vector spaces
- Determinants
- Eigenvalues
- Linear algebra software

0.2. Statistics and Probability

- Basic probability
- Random variables
- Discrete and continuous
- Confidence intervals

1. Algorithms, Computability, and Complexity

- Basic Analysis and Algorithmic Methods
- Fundamentals of Automata and Computability

2. Architecture and Organization

- Digital Logic, Digital Systems, and Representation of Data on the Machine Level
- Machine Organization on the Assembly Level
- Architecture of Memory System, Interfacing and Communication

3. Discrete Structures

- Basic Logic, Sets, Relations, and Functions
- Proof Techniques, and Basics of Counting
- Graphs, Trees, and Discrete Probability

4. Information Management

- Database Systems and Data Modeling

5. Networking and Communications

- Introduction to Networking and Communications
- Reliable Data Delivery, Routing and Forwarding, and Resource Allocation

6. Operating Systems

- Operating System Principles
- Concurrency, Scheduling, Dispatch, Memory Management, Security and Protection of OS

7. Programming Languages

- Object-Oriented Programming, and Functional Programming
- Basic Type Systems
- Program Representation, Language Translation and Execution

8. Software Development Fundamentals

- Algorithms, Design, and Development Methods
- Fundamental Programming Concepts and Data Structures

9. Software Engineering

- Software Processes, Software Management, and Requirements Engineering
- Software Design, Construction, Verification and Validation